

Aim:

Creation of single page application using angular Compounds and data binding.

Description:

The web application is an Angular SPA using components and two-way data binding, enabling dynamic UI updates, conditional rendering, reusable modules, and smooth, responsive interactions without page reloads for maintainable front-end development.

Angular Concepts Used:

- **Angular Components** – Modular building blocks to create reusable UI sections.
- **Data Binding** – Two-way binding (`[(ngModel)]`) and one-way binding (`{{ }}`) to synchronize data between UI and component.
- **Directives** – `*ngFor` for looping through datasets and `*ngIf` for conditional rendering.
- **Event Binding** – Handling user interactions such as clicks with `(click)` events.
- **Property Binding** – Dynamically updating element properties from component data.
- **FormsModule** – Capturing and validating user inputs using Angular forms.
- **Local Storage** – Storing and retrieving data in the browser for persistence.
- **Lifecycle Hooks** – Using `ngOnInit()` to initialize data when components load.
- **Asynchronous Data Handling** – Managing data updates and user interactions dynamically.
- **Error Handling** – Managing errors in user input and application logic gracefully.

Code:**App.ts**

```
import { Component, Inject, PLATFORM_ID, OnInit } from '@angular/core';
import { CommonModule, isPlatformBrowser } from '@angular/common';
import { FormsModule } from '@angular/forms';
```

```
@Component({
  selector: 'app-root',
  standalone: true,
  imports: [CommonModule, FormsModule],
  templateUrl: './app.html',
  styleUrls: ['./app.css']
})
export class App implements OnInit {
  moods: { label: string }[] = [
    { label: 'Happy' },
    { label: 'Sad' },
    { label: 'Excited' },
    { label: 'Angry' },
    { label: 'Relaxed' }
  ];
  selectedMood: string = '';
  savedMood: { label: string } | null = null;
```

```

isBrowser = false;
constructor(@Inject(PLATFORM_ID) private platformId: object) {
  this.isBrowser = isPlatformBrowser(this.platformId);
}
ngOnInit() {
  if (this.isBrowser) {
    const stored = localStorage.getItem('dailyMood');
    if (stored) {
      this.savedMood = JSON.parse(stored);
    }
  }
}
saveMood() {
  if (!this.selectedMood) {
    alert('Please select your mood for today!');
    return;
  }
  const mood = this.moods.find(m => m.label === this.selectedMood)!;
  if (this.isBrowser) {
    localStorage.setItem('dailyMood', JSON.stringify(mood));
    this.savedMood = mood;
    this.selectedMood = "";
  }
}
clearMood() {
  if (this.isBrowser) {
    localStorage.removeItem('dailyMood');
    this.savedMood = null;
  }
}
}

```

App.html

```

<div class="container">
  <h1>Daily Mood Tracker</h1>

  <div *ngIf="!savedMood">
    <label for="moodSelect">How are you feeling today?</label>
    <select id="moodSelect" [(ngModel)]="selectedMood">
      <option value="" disabled selected>Select your mood</option>
      <option *ngFor="let mood of moods" [value]="mood.label">
        {{ mood.label }}
      </option>
    </select>
    <button (click)="saveMood()">Save Mood</button>
  </div>

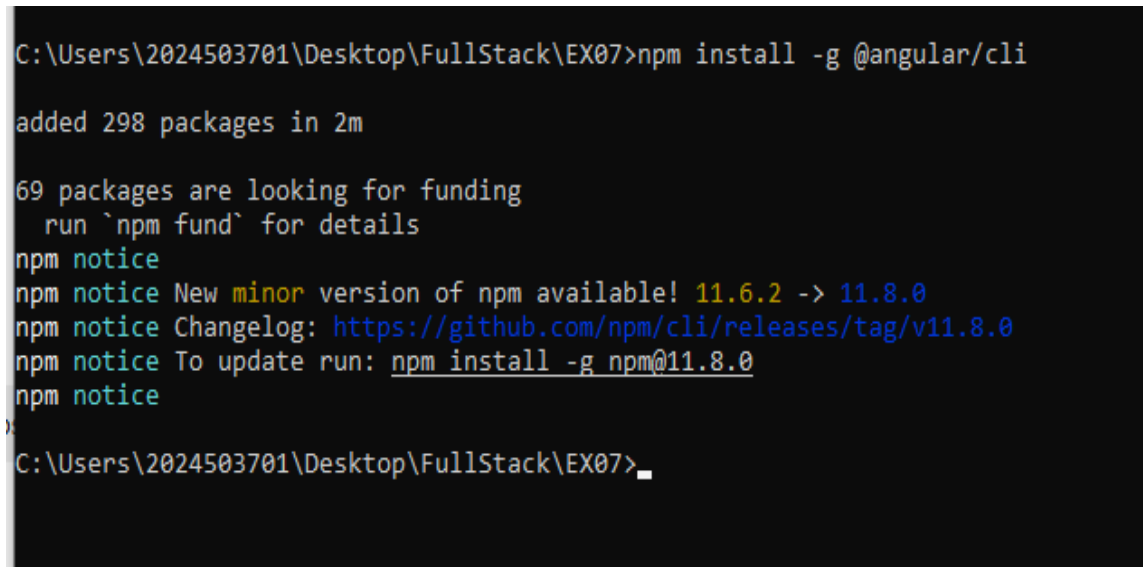
```

```
<div *ngIf="savedMood" class="mood-display">
  <h2>Today's Mood: {{ savedMood.label }}</h2>
  <button class="clear-btn" (click)="clearMood()">Reset Mood</button>
</div>
</div>
```

App.css:

```
body {
  font-family: Arial, sans-serif;
  text-align: center;
  margin: 20px;
}
.container {
  max-width: 400px;
  margin: 0 auto;
  padding: 10px;
}
select, button {
  padding: 6px;
  margin: 5px;
  border-radius: 4px;
}
.mood-display {
  margin-top: 15px;
  padding: 10px;
}
```

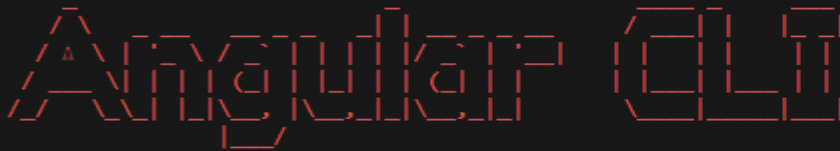
Output:

A terminal window with a dark background and light-colored text. The text shows the command 'npm install -g @angular/cli' being executed. The output indicates that 298 packages were added in 2 minutes, 69 packages are looking for funding, and a new minor version of npm (11.8.0) is available. The terminal ends with a prompt 'C:\Users\2024503701\Desktop\FullStack\EX07>' and a cursor.

```
C:\Users\2024503701\Desktop\FullStack\EX07>npm install -g @angular/cli
added 298 packages in 2m
69 packages are looking for funding
  run `npm fund` for details
npm notice
npm notice New minor version of npm available! 11.6.2 -> 11.8.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.8.0
npm notice To update run: npm install -g npm@11.8.0
npm notice
C:\Users\2024503701\Desktop\FullStack\EX07>_
```

Figure(a) Installation of Angular

```
C:\Users\2024503701\Desktop\FullStack\EX07\my-first-app>ng version
```



```
Angular CLI      : 21.1.2  
Angular         : 21.1.2  
Node.js        : 24.12.0  
Package Manager : npm 11.6.2  
Operating System : win32 x64
```

Package	Installed Version	Requested Version
@angular/build	21.1.2	^21.1.2
@angular/cli	21.1.2	^21.1.2
@angular/common	21.1.2	^21.1.0
@angular/compiler	21.1.2	^21.1.0
@angular/compiler-cli	21.1.2	^21.1.0
@angular/core	21.1.2	^21.1.0
@angular/forms	21.1.2	^21.1.0
@angular/platform-browser	21.1.2	^21.1.0
@angular/platform-server	21.1.2	^21.1.0
@angular/router	21.1.2	^21.1.0
@angular/ssr	21.1.2	^21.1.2
rxjs	7.8.2	~7.8.0
typescript	5.9.3	~5.9.2
vitest	4.0.18	^4.0.8

Figure(b)Ng version

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Initial chunk files | Names      | Raw size  
main.js           | main      | 9.41 kB |
```

Server bundles

```
Initial chunk files | Names      | Raw size  
main.server.mjs    | main.server | 10.68 kB |
```

Application bundle generation complete. [0.264 seconds] - 2026-02-02T05:55:53.942Z

Page reload sent to client(s).



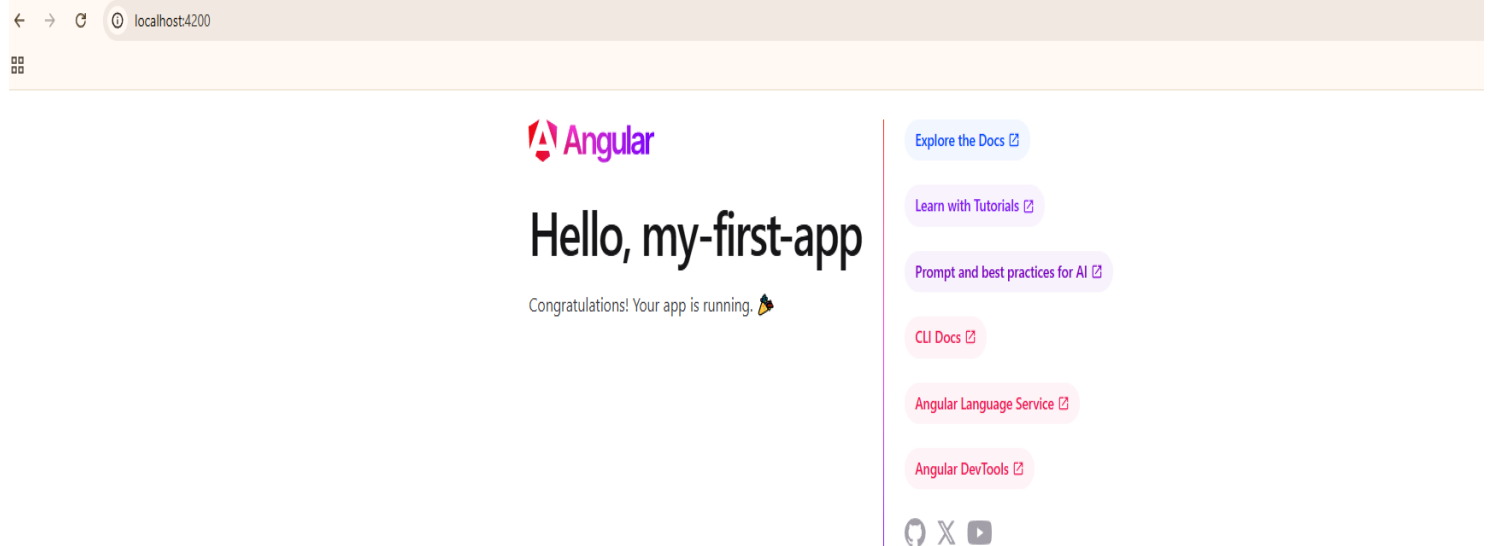
Figure(c)Running angular

```

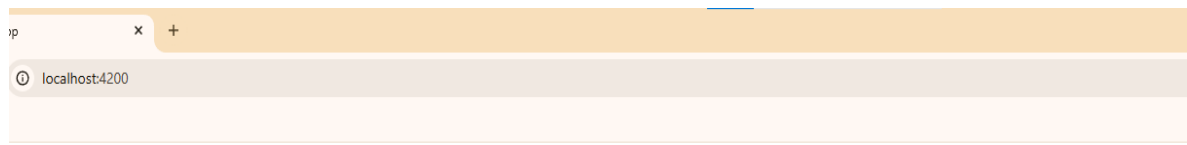
/ Do you want to enable Server-Side Rendering (SSR) and Static Site Generation (SSG/Prerendering)? Yes
/ Which AI tools do you want to configure with Angular best practices? https://angular.dev/ai/develop-with-ai None
CREATE my-first-app/angular.json (2154 bytes)
CREATE my-first-app/package.json (1196 bytes)
CREATE my-first-app/README.md (1522 bytes)
CREATE my-first-app/tsconfig.json (990 bytes)
CREATE my-first-app/.editorconfig (331 bytes)
CREATE my-first-app/.gitignore (666 bytes)
CREATE my-first-app/tsconfig.app.json (464 bytes)
CREATE my-first-app/tsconfig.spec.json (456 bytes)
CREATE my-first-app/.vscode/extensions.json (134 bytes)
CREATE my-first-app/.vscode/launch.json (490 bytes)
CREATE my-first-app/.vscode/mcp.json (188 bytes)
CREATE my-first-app/.vscode/tasks.json (1020 bytes)
CREATE my-first-app/src/main.ts (228 bytes)
CREATE my-first-app/src/index.html (309 bytes)
CREATE my-first-app/src/styles.css (81 bytes)
CREATE my-first-app/src/main.server.ts (300 bytes)
CREATE my-first-app/src/server.ts (1677 bytes)
CREATE my-first-app/src/app/app.spec.ts (702 bytes)
CREATE my-first-app/src/app/app.ts (306 bytes)
CREATE my-first-app/src/app/app.css (0 bytes)
CREATE my-first-app/src/app/app.html (20446 bytes)
CREATE my-first-app/src/app/app.config.ts (453 bytes)
CREATE my-first-app/src/app/app.routes.ts (80 bytes)
CREATE my-first-app/src/app/app.config.server.ts (438 bytes)
CREATE my-first-app/src/app/app.routes.server.ts (174 bytes)
CREATE my-first-app/public/favicon.ico (15086 bytes)
- Installing packages (npm)...

```

Figure(d) Creating new Project



Figure(e) Localhost:4200



Daily Mood Tracker

Today's Mood: Happy

Reset Mood

Figure(f) Output of File

Result:

Thus, the Single Page Application was successfully implemented using Angular components and data binding. The application dynamically displays user-selected data in the browser, providing smooth interaction and real-time UI updates without page reloads.